

The Vault

Category: Forensics

Subcategory: Firmware / S-Record Analysis

Difficulty: Easy

File: EP7PR1002RN0000.shex

Flag: OBV{7F2A_4C91_BLACK_VAULT}

Objective

Identify the Motorola S-Record file, locate the hexadecimal representation of OBV{, and decode the hidden S3 payload.

Intended Solve

Identify the file: file EP7PR1002RN0000.shex

Expected: EP7PR1002RN0000.shex: Motorola S-Record; binary data in text format

Since the flag format is OBV{...}, convert the known prefix to ASCII hex:

OBV{ → 4F42567B

Search for it: grep -in '4F42567B' EP7PR1002RN0000.shex

This reveals:

S31F00C6A4284F42567B374632415F344339315F424C41434B5F5641554C547D
98

Parse the S3 record:

S3 | 1F | 00C6A428 | 4F42567B...547D | 98

1F = 31 bytes total, consisting of 4 address bytes, 26 payload bytes, and 1 checksum byte.

Decode the payload:

```
echo '4F42567B374632415F344339315F424C41434B5F5641554C547D' | xxd -r  
-p
```

Result: OBV{7F2A_4C91_BLACK_VAULT}

Note:

GNU objcopy is not the intended route because this artifact contains custom S0 metadata that causes objcopy to reject the file. Direct S-Record inspection is the intended solve.